Numerical Computation of the Ignorance Gap for the Radar Beam-Search Case Study

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Abstract

We numerically compute the ignorance-gap distribution $IG(x)$ for the finite-POMDP radar beam-search case study of `AdaptiveDesign.tex` §9, on a single-thread laptop, using exact enumeration and no Monte Carlo approximation. The computation produces closed-form nats values for (a) the oracle value $V_{\text{oracle}}(x)$ for every cell, (b) the fixed-design optimum V_{fixed} with explicit optimal sensor sequence $s^\star$, (c) the per-cell fixed-design value $\Phi(x, s^\star, c)$, and (d) the per-cell ignorance gap $IG(x)$ in both the narrow-beam and wide-beam regimes. The document also clarifies one conceptual subtlety: the terminal reward used in the discrete-state numerical problem is the mutual information $I(z; y_{1:N})$, not a literal Fisher-information log-determinant, and the two agree only in the continuous-state Gaussian-posterior limit.

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1 Purpose and scope

The companion document `AdaptiveDesign.tex` develops the ignorance gap

$$IG(\boldsymbol{x}) = V_{\text{oracle}}(\boldsymbol{x}) - V_{\text{fixed}}(\boldsymbol{x}; s^\star)$$

as an EVPI upper bound on the value of any adaptive policy. Section 9 of that document analyzes the radar beam-search problem in two regimes—narrow beam and wide beam—and derives closed-form predictions for $\mathbb{E}[IG]$ in the idealized $(p_{\text{D}}, p_{\text{FA}}) \to (1, 0)$ limit.

This note replaces the idealized estimates with *exact finite-noise numerics* by enumerating all finite POMDP quantities directly. The problem size ($K = 16$ states, $M = 16$ actions, $|\mathcal{Y}| = 2$ observations, horizon $N = 4$) is small enough that no approximation, sampling, or solver beyond exhaustive enumeration is needed. The intent is to (i) verify the qualitative predictions, (ii) quantify the finite-noise correction, (iii) expose the full distribution of $IG(x)$, and (iv) surface one conceptual subtlety that had been glossed in the main document.

2 Discrete-state reformulation of the terminal reward

The main document defines the terminal reward as

$$\Phi(z, \theta_{1:N}, c) \;=\; \ln K - H(z \mid y_{1:N})$$

and notes that this equals $\ln\det \boldsymbol{J}_N$ "up to an additive constant under the categorical's softmax parametrization." That equivalence is loose in the discrete-state setting: there is no continuous parameter x to differentiate, so Fisher information as a Hessian is ill-defined. The honest discrete analogue is the *mutual information*

$$\Phi(z, s, c) \;=\; I(z;\, y_{1:N} \mid s, c) \;=\; \ln K - \mathbb{E}_{y|z,s,c}[H(z \mid y_{1:N}, s, c)], \tag{1}$$

which is well-defined for any finite z-alphabet, matches the "information-gain" reading of Φ, and coincides with $\ln\det \boldsymbol{J}_N$ only in the continuous-state Gaussian-posterior limit (where a softmax-logit Gaussian approximation of the posterior becomes exact).

For the numerical work we use Φ as defined in (1) throughout. No Hessian or log-determinant is ever computed; everything reduces to probabilities and entropies on a finite alphabet.

3 Problem parameters and beam model

State. $K = 16$ cells equally spaced on a ring: $\varphi(z) = 2\pi(z-1)/K$ for $z \in \{1, \ldots, K\}$. Prior $p_0(z) = 1/K$.

Actions. $M = 16$ pointings: $\theta_a = \varphi(a)$ for $a \in \{1, \ldots, K\}$. The action at step k is an index $s_k \in \{1, \ldots, K\}$ denoting the cell the beam is centered on.

Observations. Binary $y_k \in \{0, 1\}$ (detection / miss) with Bernoulli likelihood

$$p(y_k = 1 \mid z, s_k) = \begin{cases} p_{\mathrm{D}} & \text{if } z \text{ lies in the beam centered at cell } s_k, \\ p_{\mathrm{FA}} & \text{otherwise.} \end{cases} \tag{2}$$

Beam model (top-hat). The geometry c is summarized by a single integer m = number of cells covered by the main lobe. Let $d(a, z) = \min(\mathrm{mod}(z-a, K),\, \mathrm{mod}(a-z, K))$ be the angular distance (in cells) between pointing a and cell z. Then

$$z \text{ is in beam centered at } a \iff d(a, z) \le \lfloor (m-1)/2 \rfloor. \tag{3}$$

Two regimes are reported:

- **Narrow-beam:** $m = 1$. Beam at cell a covers only cell a.
- **Wide-beam:** $m = K/2 = 8$. The even-m convention in our code yields a beam covering 7 cells (the target cell plus 3 on each side). This is an order-one rounding of $K/2$; qualitative behavior matches the wide-beam regime of the main document.

Probabilities. $p_{\mathrm{D}} = 0.9$, $p_{\mathrm{FA}} = 0.05$. These are finite; the $(p_{\mathrm{D}}, p_{\mathrm{FA}}) \to (1, 0)$ closed-form predictions of the main document are upper-bound limits.

4 Posterior and entropy formulas

Given a sensor sequence $s_{1:N}$ and observations $y_{1:N}$, the posterior on z is

$$b(z \mid y_{1:N}, s_{1:N}) \;=\; \frac{p_0(z) \prod_{k=1}^{N} p(y_k \mid z, s_k)}{\sum_{z'} p_0(z') \prod_{k=1}^{N} p(y_k \mid z', s_k)}. \tag{4}$$

Its Shannon entropy (nats) is

$$H(z \mid y_{1:N}, s_{1:N}) = -\sum_{z=1}^{K} b(z \mid y_{1:N}, s_{1:N}) \ln b(z \mid y_{1:N}, s_{1:N}). \tag{5}$$

The state-conditional value of Φ is the observation-averaged information gain:

$$\Phi(x, s_{1:N}, c) = \sum_{y_{1:N} \in \{0,1\}^N} p(y_{1:N} \mid x, s_{1:N}) \left[\ln K - H(z \mid y_{1:N}, s_{1:N})\right] \tag{6}$$

with

$$p(y_{1:N} \mid x, s_{1:N}) = \prod_{k=1}^{N} p(y_k \mid x, s_k). \tag{7}$$

The sum in (6) is over $2^N = 16$ terms (a direct enumeration); the posterior in (4) sums over $K = 16$ terms. Per $(x, s_{1:N})$ evaluation: $16 \times (K \cdot N + K) \approx 16 \times 80 = 1280$ elementary operations.

5 Oracle value $V_{\text{oracle}}(x)$

5.1 Definition

$$V_{\text{oracle}}(x) = \max_{s_{1:N}} \Phi(x, s_{1:N}, c).$$

5.2 Algorithm

For each of the $K = 16$ values of x, enumerate all $K^N = 65{,}536$ sensor sequences and retain the maximum. Cost: $16 \times 65{,}536 \times 1280 \approx 1.3 \times 10^9$ elementary operations. Julia single-threaded runtime: a few seconds.

5.3 Symmetry prediction

The problem has a dihedral D_K symmetry (ring cyclic rotations plus one reflection) under the uniform prior and reflection-symmetric beam. Consequently $V_{\text{oracle}}(x)$ is *exactly* constant across x: if $s^\star$ is oracle-optimal for $x = z_0$, then $\sigma \cdot s^\star$ is oracle-optimal for $\sigma \cdot z_0$ with the same value.

5.4 Numerical result

x	$V_{\text{oracle}}(x)$	$s^\star(x)$
1, …, 16	2.579634	narrow: $[x, x, x, x]$
1, …, 16	1.923462	wide: (see text)

Table 1: Numerical $V_{\text{oracle}}(x)$ in nats, for both regimes. Constant to 15 decimal places over x—the residual variation is machine ϵ.

Narrow regime. Oracle strategy is trivially $s^\star(x) = (x, x, x, x)$: point at the known target every step. Per-step on-target Bernoulli is $p_{\text{D}} = 0.9$ (vs. $p_{\text{FA}} = 0.05$ for every other cell).

Wide regime. Oracle strategy is non-trivial: for $x = 1$, numerical $s^\star = [3, 14, 14, 4]$. Each of these pointings places cell 1 inside its 7-cell beam, so every measurement is on-target from cell 1's perspective; but the three distinct pointings exclude complementary non-1 cells. Specifically,

$$\begin{aligned} \text{beam at 3 excludes } &\{7, 8, 9, 10, 11, 12, 13\}, \\ \text{beam at 14 excludes } &\{2, 3, 4, 5, 6, 7, 8, 9, 10\}, \\ \text{beam at 4 excludes } &\{8, 9, 10, 11, 12, 13, 14, 15, 16\}. \end{aligned}$$

The union covers all cells except cell 1, so three *distinct* detections together triangulate the target. The oracle exploits the geometric overlap structure—a capability that narrow-beam oracle (single-cell beams) cannot use.

5.5 Why $V_{\text{oracle}}^{\text{narrow}} > V_{\text{oracle}}^{\text{wide}}$

Narrow beams separate a single cell from all others at $p_{\text{D}}/p_{\text{FA}} = 18{:}1$ per measurement. Wide beams separate $m = 7$ cells from the $K - m = 9$ remaining at the same per-shot ratio, but the oracle can no longer isolate a single cell with one measurement because seven cells share the "in-beam" label. Narrow per-step information is higher; oracle is stronger. Specifically $V_{\text{oracle}}^{\text{narrow}} = 2.580$ nats vs. $V_{\text{oracle}}^{\text{wide}} = 1.923$ nats, both bounded above by the prior entropy $\ln K = 2.773$ nats.

6 Fixed-design value V_{fixed}

6.1 Definition

$$V_{\text{fixed}} = \max_{s_{1:N}} \frac{1}{K} \sum_{x=1}^{K} \Phi(x, s_{1:N}, c), \qquad s^\star = \arg\max_{s_{1:N}} \frac{1}{K} \sum_{x=1}^{K} \Phi(x, s_{1:N}, c).$$

6.2 Algorithm

Enumerate all $K^N = 65{,}536$ sensor sequences; for each, average Φ over the uniform prior and track the maximum. Total per-sequence cost includes the inner $K = 16$ state loop: $65{,}536 \times 16 \times 1280 \approx 1.3 \times 10^9$ operations. Julia runtime: a few seconds.

6.3 Numerical result

regime	$s^\star$	V_{fixed} (nats)
narrow	$[1, 13, 9, 7]$ (any 4 distinct cells, up to D_K orbit)	0.498683
wide	$[1, 3, 13, 15]$ (4 spread pointings)	1.410526

Table 2: Fixed-design optimum, exhaustive enumeration.

6.4 Per-cell breakdown of $\Phi(x, s^\star, c)$

Unlike $V_{\text{oracle}}(x)$, the fixed-design conditional value $\Phi(x, s^\star, c)$ is *not* constant over x because a deterministic $s^\star$ breaks the ring symmetry. Table 3 shows the per-cell breakdown for both regimes.

x	$\Phi(x, s^\star)$ narrow	$\Phi(x, s^\star)$ wide
1	0.976697	1.540807
2	0.339346	1.540807
3	0.339346	1.431264
4	0.339346	1.431264
5	0.339346	1.354544
6	0.339346	1.354544
7	0.976697	1.164713
8	0.339346	1.164713
9	0.976697	1.164713
10	0.339346	1.354544
11	0.339346	1.354544
12	0.339346	1.431264
13	0.976697	1.431264
14	0.339346	1.540807
15	0.339346	1.540807
16	0.339346	1.767821
avg	0.498683	1.410526

Table 3: Conditional fixed-design value $\Phi(x, s^\star, c)$ for each x. Narrow: 2 distinct values (probed / unprobed). Wide: 5 distinct values (the cell's position relative to $s^\star = [1, 3, 13, 15]$). Averages match V_{fixed} exactly.

Narrow regime interpretation. With $s^\star = [1, 13, 9, 7]$, the four probed cells each see one on-target beam (which gives a detection with probability $p_{\text{D}} = 0.9$), leading to high information about whether z equals that cell. The 12 unprobed cells see four off-target beams and the information content is much lower.

$$\Phi(\text{probed}) = 0.976697 \text{ nats}$$
$$\Phi(\text{unprobed}) = 0.339346 \text{ nats}$$

The prior-weighted average is $V_{\text{fixed}} = (4/16)(0.977) + (12/16)(0.339) = 0.499$ nats.

Wide regime interpretation. With $s^\star = [1, 3, 13, 15]$ (four spread pointings) and $m = 7$-cell beams, different cells see different numbers of on-target beams:

- Cell 16 lies in *all four* beams (closest to all four pointings by ring wraparound) and receives the highest $\Phi = 1.768$.
- Cells 1, 2, 14, 15 lie in three beams and have $\Phi = 1.541$.
- Cells 3, 4, 12, 13 lie in two beams and have $\Phi = 1.431$.
- Cells 5, 6, 10, 11 lie in one beam and have $\Phi = 1.355$.
- Cells 7, 8, 9 (the "back" of the ring, between pointings 3 and 13) lie in *zero* beams: $\Phi = 1.165$.

The five-level structure directly reflects the in-beam count, independent of cell identity.

7 Ignorance gap $IG(x)$

7.1 Definition and computation

$$IG(x) = V_{\text{oracle}}(x) - \Phi(x, s^\star, c).$$

Since $V_{\text{oracle}}(x)$ is constant over x (exact symmetry) and $\Phi(x, s^\star, c)$ has been tabulated per x, this is a direct subtraction.

7.2 Narrow regime

x	V_{oracle}	$\Phi(x, s^\star)$	$IG(x)$
1	2.579634	0.976697	1.602937
2	2.579634	0.339346	2.240288
3	2.579634	0.339346	2.240288
4	2.579634	0.339346	2.240288
5	2.579634	0.339346	2.240288
6	2.579634	0.339346	2.240288
7	2.579634	0.976697	1.602937
8	2.579634	0.339346	2.240288
9	2.579634	0.976697	1.602937
10	2.579634	0.339346	2.240288
11	2.579634	0.339346	2.240288
12	2.579634	0.339346	2.240288
13	2.579634	0.976697	1.602937
14	2.579634	0.339346	2.240288
15	2.579634	0.339346	2.240288
16	2.579634	0.339346	2.240288
mean			$\mathbb{E}[IG] = 2.080950$
std			0.285032
distinct values			2

Table 4: Narrow-regime $IG(x)$ table. The 4 probed cells (1, 7, 9, 13) have $IG = 1.603$; the 12 unprobed cells have $IG = 2.240$.

The two-value distribution in the narrow regime is a direct consequence of the 4-vs-12 partition created by $s^\star$. Cells inside $s^\star$ have lower IG because the fixed design does give them one on-target probe, reducing the gap to the oracle. Cells outside $s^\star$ are treated identically (all-off-target beams) and hence have identical IG.

7.3 Wide regime

Across the wide regime the gap varies by a factor of nearly 5× (from 0.156 to 0.759), with cell 16 the most-covered and cells 7–9 the "shadow" of the spread pointings $[1, 3, 13, 15]$.

8 Why $IG(x)$ is not constant across x

At first glance the non-constancy is surprising: the prior is uniform, the ring is D_K-symmetric, the beam pattern is rotation-equivariant—so shouldn't every cell look equivalent?

The resolution is that the oracle and the fixed design treat the symmetry group differently.

Oracle. By definition, $V_{\text{oracle}}(x) = \max_s \Phi(x, s, c)$. For each x, the oracle *re-optimizes* and picks the best sensor sequence for that particular x. Under the group action $\sigma \in D_K$, the optimal sensor sequence for $\sigma \cdot x$ is $\sigma \cdot s^\star(x)$, and $\Phi(\sigma \cdot x, \sigma \cdot s^\star(x), \sigma \cdot c) = \Phi(x, s^\star(x), c)$ by equivariance. Hence $V_{\text{oracle}}(x)$ is truly constant over x.

Fixed design. The best fixed sensor sequence $s^\star_{\text{fixed}}$ is chosen once and used for every x. For a single deterministic $s^\star$, the value $\Phi(x, s^\star, c)$ under the group action becomes $\Phi(\sigma \cdot x, s^\star, c)$, which does *not* equal $\Phi(x, s^\star, c)$ unless $s^\star$ is σ-invariant. No single $s^\star$ is invariant under the full

x	V_{oracle}	$\Phi(x, s^\star)$	$IG(x)$
1	1.923462	1.540807	0.382655
2	1.923462	1.540807	0.382655
3	1.923462	1.431264	0.492197
4	1.923462	1.431264	0.492197
5	1.923462	1.354544	0.568918
6	1.923462	1.354544	0.568918
7	1.923462	1.164713	0.758748
8	1.923462	1.164713	0.758748
9	1.923462	1.164713	0.758748
10	1.923462	1.354544	0.568918
11	1.923462	1.354544	0.568918
12	1.923462	1.431264	0.492197
13	1.923462	1.431264	0.492197
14	1.923462	1.540807	0.382655
15	1.923462	1.540807	0.382655
16	1.923462	1.767821	0.155641
mean			$\mathbb{E}[IG] = 0.512935$
std			0.161468
distinct values			5

Table 5: Wide-regime $IG(x)$ table. Five distinct values, indexed by how many of the four $s^\star$ beams cover cell x.

D_K orbit, so $\Phi(x, s^\star, c)$ must vary across the orbit—and by size of its stabilizer, it takes on a number of distinct values equal to K divided by the orbit length of $s^\star$.

Consequence. The deterministic fixed-design value decomposes the K cells according to their position relative to $s^\star$:

- Narrow regime: 4 cells in $s^\star$, 12 outside. Two-valued IG.
- Wide regime: 5 classes depending on how many $s^\star$-beams cover each cell (0, 1, 2, 3, or 4 beams). Five-valued IG.

The prior-expected IG is

$$\mathbb{E}[IG] = V_{\text{oracle}} - V_{\text{fixed}} = V_{\text{oracle}} - \mathbb{E}_x[\Phi(x, s^\star, c)],$$

which is a single scalar, but the orbit-dependent distribution of $\Phi(x, s^\star, c)$ around that mean gives the per-cell spread of $IG(x)$.

Orbit averaging would restore constancy. A *randomized* fixed design that applies $s^\star$ with uniform probability over its D_K orbit would give $\mathbb{E}_\sigma[\Phi(x, \sigma \cdot s^\star, c)]$ constant in x by symmetry. The deterministic design sacrifices this for the best single-realization value. In a sense, the variability of $IG(x)$ is the price of deterministic commitment.

9 Comparison with the closed-form predictions

9.1 Ideal-limit predictions

The main document's §9 derives, under $(p_{\mathrm{D}}, p_{\mathrm{FA}}) \to (1, 0)$:

$$V_{\mathrm{oracle}} \approx \ln K = 2.773 \text{ nats}, \tag{8}$$

$$V_{\mathrm{fixed}}^{\mathrm{narrow}} \approx \ln K - \frac{K-N}{K} \ln(K-N) = 0.909 \text{ nats}, \tag{9}$$

$$V_{\mathrm{fixed}}^{\mathrm{wide}} \approx \ln(2N) = 2.079 \text{ nats}, \tag{10}$$

$$\mathbb{E}[IG]^{\mathrm{narrow}} \approx \frac{K-N}{K} \ln(K-N) = 1.864 \text{ nats}, \tag{11}$$

$$\mathbb{E}[IG]^{\mathrm{wide}} \approx \ln\big(K/(2N)\big) = 0.693 \text{ nats}. \tag{12}$$

9.2 Ideal vs. finite-noise

quantity	ideal $(p_{\mathrm{D}} \to 1, p_{\mathrm{FA}} \to 0)$	finite $p_{\mathrm{D}} = 0.9, p_{\mathrm{FA}} = 0.05$	ratio
$V_{\mathrm{oracle}}^{\mathrm{narrow}}$	2.773	2.580	0.93
$V_{\mathrm{fixed}}^{\mathrm{narrow}}$	0.909	0.499	0.55
$\mathbb{E}[IG]^{\mathrm{narrow}}$	1.864	2.081	1.12
$V_{\mathrm{oracle}}^{\mathrm{wide}}$	2.773	1.923	0.69
$V_{\mathrm{fixed}}^{\mathrm{wide}}$	2.079	1.411	0.68
$\mathbb{E}[IG]^{\mathrm{wide}}$	0.693	0.513	0.74

Table 6: Idealized closed-form predictions vs. exact finite-noise numerics.

9.3 Interpretation

- *Narrow regime.* $\mathbb{E}[IG]$ is *larger* at finite noise than predicted (2.081 vs. 1.864). The reason: both V_{oracle} and V_{fixed} shrink as noise increases, but V_{fixed} shrinks more aggressively (factor 0.55) than V_{oracle} (factor 0.93). The oracle can always point at the target and extract information efficiently even at finite noise; the fixed design wastes most of its budget on cells that aren't the target, and each off-target beam gives a weak Bernoulli(p_{FA}) whose information content degrades with increasing p_{FA}. The gap *widens* as noise grows.

- *Wide regime.* All three quantities shrink by roughly the same factor (≈ 0.7). The gap is dominated by the coverage structure, which degrades oracle and fixed-design roughly symmetrically as noise increases.

The qualitative structure predicted by the main document—$\mathbb{E}[IG]^{\mathrm{narrow}} > \mathbb{E}[IG]^{\mathrm{wide}}$—holds unambiguously: $2.08 > 0.51$. The numerical margin is even larger than the ideal-limit prediction.

10 Summary of IG statistics

The coefficient of variation std/$\mathbb{E}[IG]$ is larger for the wide regime (0.31 vs. 0.14), consistent with the richer 5-level structure: even though $\mathbb{E}[IG]$ is smaller, its distribution across cells is *more* heterogeneous in relative terms. If one were to deploy an adaptive policy, the cells with the highest $IG(x)$ (indices 7, 8, 9 in the wide regime) would be the ones where adaptation could in principle help the most.

statistic	narrow	wide
$\mathbb{E}[IG]$	2.080950	0.512935
$\mathrm{std}[IG]$	0.285032	0.161468
$\min[IG]$	1.602937	0.155641
$\max[IG]$	2.240288	0.758748
max / min	1.40	4.87
distinct values	2	5
CV = std/$\mathbb{E}$	0.14	0.31

Table 7: Distributional statistics of $IG(x)$.

11 Computational cost summary

All reported quantities come from exact enumeration, no sampling. Per-regime:

- V_{oracle}: $K \cdot K^N \cdot 2^N \cdot (KN + K)$ elementary operations $= 16 \cdot 65{,}536 \cdot 16 \cdot 80 \approx 1.3 \times 10^9$.
- V_{fixed}: identical count.

Total for both regimes: $\approx 5 \times 10^9$ elementary operations. Julia 1.12 single-threaded completes both in well under a minute on a standard laptop. RAM footprint: under 20 MB (the largest array is one posterior vector of length $K = 16$ per outcome, recomputed on the fly).

12 Conceptual reading of the results

12.1 The oracle is symmetric; the deterministic fixed design is not

The equality $V_{\text{oracle}}(x) = \text{const}$ is an exact symmetry fact. The non-constancy of $IG(x)$ comes entirely from $\Phi(x, s^\star, c)$, the per-x fixed-design value, which depends on the x-orbit position relative to a single chosen $s^\star$. The narrow regime has a simple binary decomposition (probed vs. unprobed); the wide regime has a richer 5-level structure determined by cell-to-beam overlap count. Both are fully predicted by the orbit structure.

12.2 Finite noise widens the narrow-regime gap

The ideal-limit prediction $\mathbb{E}[IG]^{\text{narrow}} \approx 1.86$ nats becomes 2.08 nats at finite noise because V_{fixed} shrinks faster than V_{oracle}. This is a second-order effect not visible in the idealized analysis, but structurally expected: the fixed design's off-target beams have p_{FA}-dominated likelihoods that degrade sharply as p_{FA} grows from 0.

12.3 $\mathbb{E}[IG]$ as a diagnostic, not a promise

In the narrow regime, $\mathbb{E}[IG] = 2.08$ nats is substantial (EVPI upper bound of $\sim 2.5\times$ uncertainty-volume reduction), but as the main document argues, adaptation cannot realize any of this in a narrow-beam setting: with single-cell beams, adaptive and fixed designs both visit N cells in N steps, in the same expected order under the uniform prior. Section 9 of `AdaptiveDesign.tex` predicts $V^{\text{narrow}}_{\text{adaptive}} = V^{\text{narrow}}_{\text{fixed}} = 0.499$ nats. Verifying this—and verifying the tight-EVPI result $V^{\text{wide}}_{\text{adaptive}} - V^{\text{wide}}_{\text{fixed}} = \mathbb{E}[IG]^{\text{wide}} = 0.51$ nats—requires the full finite-horizon Bellman DP, which is the next computational step.

13 Exact Bellman DP and V_{adaptive}

13.1 Algorithm

The terminal reward $\Phi = \ln K - H(z \mid y_{1:N})$ is a nonlinear functional of the terminal belief, so Sondik's piecewise-linear-convex representation does not apply directly. We instead do backward induction on the belief tree reachable from the uniform prior. The recursion is

$$W_k(b) = \max_{a \in \{1,\ldots,K\}} \sum_{y \in \{0,1\}} P(y \mid b, a) \cdot W_{k+1}\big(T(b, a, y)\big), \qquad W_{N+1}(b) = \ln K - H(b), \tag{13}$$

where $T(b, a, y)$ is the Bayesian update of b under observation y when taking action a.

At depth d from root the recursion branches by (K actions) $\times$ (2 outcomes), giving $(2K)^N = 32^4 \approx 10^6$ terminal evaluations in the worst case. Each evaluation is $O(K)$ for the posterior update and entropy. Total work: $\approx 10^7$ elementary operations per regime. Julia runtime: ~ 0.1 seconds. Memory peak: a handful of belief vectors of length K, so < 1 MB.

13.2 Numerical results

	narrow (m=1)	wide (m=$K/2$=8)
V_{oracle}	2.579634	1.923462
V_{adaptive}	0.619198	1.699767
V_{fixed}	0.498683	1.410526
$\mathbb{E}[IG]$ (EVPI bound)	2.080950	0.512935
$V_{\text{adaptive}} - V_{\text{fixed}}$ (realized gain)	0.120515	0.289241
EVPI saturation %	5.8%	56.4%

Table 8: Exact Bellman-DP results with $K = 16$, $N = 4$, $p_{\text{D}} = 0.9$, $p_{\text{FA}} = 0.05$. All numbers in nats unless noted. "EVPI saturation" is the fraction $(V_{\text{adaptive}} - V_{\text{fixed}})/\mathbb{E}[IG]$, showing how much of the ignorance-gap upper bound the optimal adaptive policy actually realizes.

13.3 Optimal policy tree (first two levels)

Narrow regime. The depth-2 head of the DP-optimal policy tree:

history	action
$\emptyset$ (root)	cell 1
$y_1 = 0$ (miss)	cell 1
$y_1 = 1$ (detect)	cell 16
$(y_1, y_2) = (0, 0)$	cell 1
$(y_1, y_2) = (0, 1)$	cell 16
$(y_1, y_2) = (1, 0)$	cell 16
$(y_1, y_2) = (1, 1)$	cell 16

The policy probes cell 1 at the first step and, if the observation was a miss, probes it *again* at step 2 (because at finite noise a single miss does not definitively rule out cell 1: the posterior on cell 1 only dropped from $1/16 = 0.0625$ to ≈ 0.007). After a detection it switches to a different cell.

Wide regime. The depth-2 head:

history	action
$\emptyset$ (root)	cell 1
$y_1 = 0$ (miss)	cell 4
$y_1 = 1$ (detect)	cell 13
$(0, 0)$	cell 4
$(0, 1)$	cell 3
$(1, 0)$	cell 3
$(1, 1)$	cell 4

The wide-beam adaptive policy shows visible binary-search structure: a detection at cell 1 (which covers cells $\{14, 15, 16, 1, 2, 3, 4\}$) narrows the support to roughly the top half of the ring; a miss narrows to the bottom half. Step 2 at cell 4 (which covers $\{1, \ldots, 7\}$) vs. cell 13 (which covers $\{10, \ldots, 16\}$) is consistent with bisecting the post-step-1 support.

13.4 Departures from idealized predictions

Both regimes diverge from the idealized predictions of `AdaptiveDesign.tex` §9 in the direction expected once finite Bernoulli noise is taken into account:

Narrow regime predicted zero adaptive benefit; actually 0.12 nats. The main document's §9 claim that "under the strictly narrow-beam restriction, $V_{\text{adaptive}} = V_{\text{fixed}}$" is exact only in the ideal $(p_{\text{D}}, p_{\text{FA}}) \to (1, 0)$ limit. At finite noise the adaptive policy does realize a small but strictly positive gain—5.8% of the EVPI ceiling. The mechanism is visible in the policy tree: after a detection the adaptive policy *switches* to a different cell (since the on-target cell is now likely resolved), whereas a pre-committed fixed schedule would continue visiting cells determined in advance. The benefit is modest because the fixed design with 4 distinct-cell pointings is *already* almost-optimally-chosen under the uniform prior, leaving adaptation little room.

Wide regime predicted $\ln 2 \approx 0.69$ nats gain; actually 0.29 nats. The §9 idealized prediction $V_{\text{adaptive}} - V_{\text{fixed}} = \ln(K/(2N)) = \ln 2$ is an upper bound assumed to be saturated when adaptive binary search can fully resolve. At finite $(p_{\text{D}}, p_{\text{FA}})$ full resolution isn't achievable in $N = 4$ noisy binary measurements (the information bound $4 \ln 2 = 2.77$ nats is not reached by either side). Adaptive achieves 1.70 nats, fixed 1.41, leaving a realized gap of 0.29 nats—56% of the EVPI. The direction matches prediction but the magnitude is reduced by noise.

EVPI tightness structure is actually reversed from the idealization. The idealized analysis suggested the wide regime had tight EVPI and the narrow regime slack EVPI. Numerically this is qualitatively preserved (56% vs. 6% saturation), but the narrow regime is *more* slack than predicted (expected 0%, got 6%, which is in the right direction), and the wide regime is *less* tight than predicted (expected 100%, got 56%). The quantitative message is that *EVPI is a genuinely loose upper bound at finite noise in all regimes we checked*, and the realized adaptive benefit is always strictly less than $\mathbb{E}[IG]$. This strengthens rather than weakens the case for running an exact DP when adaptation is being considered.

13.5 Key quantitative takeaway

At the joint $(\boldsymbol{c}, \pi)$ optimum (wide beam, adaptive policy), finite-noise numerics give

$$V^{\star}_{\text{adaptive}} - V^{\star}_{\text{fixed}} = 0.289 \text{ nats} \;\approx\; e^{0.289/2} - 1 \approx 16\% \text{ reduction in CRB volume.} \tag{14}$$

This is the *actual* dividend of adopting the full adaptive framework at this problem size—measurable, positive, but about half of what the idealized closed-form suggested. The ignorance-gap go/no-go gate ($\mathbb{E}[IG] > 1$ nat in the narrow regime) would have recommended proceeding with the DP; post-solve, we see the realized gain was much smaller but still present.

14 Joint geometry–policy optimization (Gaussian beam)

The top-hat beam of §3 is discrete in m, so gradient-based outer optimization of the geometry is not directly meaningful. We now replace the beam model with a continuously parametrized Gaussian and sweep w over a grid to map the joint $(\boldsymbol{c}, \pi)$ landscape.

14.1 Gaussian-beam likelihood

Parametrize the geometry by a single continuous scalar $w > 0$ (beamwidth):

$$G_w(\alpha) = \exp\bigl(-\alpha^2/(2w^2)\bigr), \qquad p(y_k = 1 \mid z, a; w) = p_{\mathrm{FA}} + (p_{\mathrm{D}} - p_{\mathrm{FA}})\, G_w\bigl(\theta_a - \varphi(z)\bigr), \tag{15}$$

with α wrapped to $[-\pi, \pi]$. All downstream quantities—$\Phi(x, s, w)$, $V_{\mathrm{oracle}}(w)$, $V_{\mathrm{fixed}}(w)$, $V_{\mathrm{adaptive}}(w)$—are smooth in w. We sweep w on a grid and at each value compute all three values exactly by the same algorithms as §5–7 (symmetry-reduced enumeration for oracle/fixed, backward induction for the DP).

14.2 Sweep results

The total sweep (19 grid points, $w \in [0.05, \pi]$) completed in 122 seconds on a single Julia thread.

w (rad)	V_{oracle}	V_{adaptive}	V_{fixed}	$\mathbb{E}[IG]$	realized gain	sat. %
0.05	2.5796	0.6192	0.4987	2.0810	0.1205	5.8
0.10	2.5795	0.6191	0.4986	2.0809	0.1206	5.8
0.15	2.5666	0.6176	0.4924	2.0742	0.1252	6.0
0.20	2.4792	0.6411	0.4931	1.9861	0.1480	7.5
0.25	2.3035	0.6875	0.5346	1.7689	0.1529	8.6
0.30	2.1210	0.7335	0.5889	1.5320	0.1445	9.4
0.35	1.9500	0.7704	0.6358	1.3142	0.1347	10.2
0.40	1.8023	0.8059	0.6729	1.1294	0.1330	11.8
0.50	1.5714	0.8522	0.7301	0.8414	0.1222	14.5
0.60	**1.3922**	**0.8647**	0.7571	0.6351	0.1077	17.0
0.70	1.2426	0.8532	**0.7576**	0.4850	0.0956	19.7
0.80	1.1192	0.8264	0.7480	0.3712	0.0784	21.1
1.00	0.9118	0.7470	0.6968	0.2150	0.0502	23.4
1.25	0.7041	0.6305	0.6016	0.1025	0.0290	28.3
1.50	0.5819	0.5159	0.4983	0.0836	0.0176	21.1
1.75	0.5284	0.4146	0.4034	0.1250	0.0112	9.0
2.00	0.4698	0.3310	0.3234	0.1464	0.0076	5.2
2.50	0.3415	0.2119	0.2079	0.1336	0.0041	3.0
π	0.2117	0.1237	0.1215	0.0902	0.0022	2.4

Table 9: Full sweep of the Gaussian-beam geometry parameter. Joint optima in bold: $V^\star_{\mathrm{adaptive}} = 0.865$ nats at $w = 0.60$, $V^\star_{\mathrm{fixed}} = 0.758$ nats at $w = 0.70$. The rightmost column is the EVPI saturation fraction.

14.3 Observations

Optimum beam width. V_{adaptive} peaks at $w^\star \approx 0.60$ rad. In HPBW units (FWHM of the Gaussian, $= 2w\sqrt{2\ln 2} \approx 2.355\,w$), this corresponds to HPBW$^\star \approx 1.41$ rad ≈ 3.6 cells. This is neither the narrow-beam limit (HPBW $\ll 2\pi/K$) nor the wide-beam limit (HPBW $\sim \pi$). The joint optimum lies in an intermediate regime where each measurement resolves a small neighborhood of cells well enough that four measurements can collectively localize the target.

Why $\mathbf{V_{oracle}}$ decreases monotonically in w. The oracle knows x and points the beam directly at it. Narrower beams give larger $p(y = 1 \mid z = x) - p(y = 1 \mid z = x')$ gap, hence more information per measurement. No finite-noise ceiling binds: as $w \to 0$, $V_{\text{oracle}} \to V_{\text{oracle}}^{\text{narrow}} = 2.58$ nats (recovering the Section 5 value for the $m = 1$ top-hat).

Why $\mathbf{V_{adaptive}}$ and $\mathbf{V_{fixed}}$ have interior maxima. At very narrow w, per-measurement information is high but *only for cells near the pointing.* With $N = 4$ pointings across $K = 16$ cells, fixed and adaptive both miss most of the prior mass most of the time. At very wide w, per-measurement information degrades and the best binary partition is weakly diagnostic. The interior optimum $w^\star \approx 0.6$ balances the two.

Where the EVPI saturation is highest. Saturation peaks at $w \approx 1.25$ rad (28.3%), far from the $\mathrm{V}_{\text{adaptive}}$ optimum. At large w the EVPI itself is small (around 0.1 nats) and adaptation captures a relatively large fraction of a small gap. At the $\mathrm{V}_{\text{adaptive}}$ optimum ($w = 0.6$), saturation is only 17%—adaptation captures a small fraction of a moderately-sized EVPI, but the *absolute* realized gain at $w = 0.6$ (0.108 nats) is closer to the maximum of the realized-gain curve (0.153 at $w = 0.25$) than the saturation percentage alone would suggest.

Joint optima for adaptive vs. fixed differ slightly. V_{fixed} peaks at $w = 0.70$, V_{adaptive} at $w = 0.60$. The adaptive optimum prefers a slightly narrower beam because it can exploit the higher per-measurement information by routing each measurement based on earlier outcomes; the fixed design, committed to a non-adaptive schedule, benefits from a slightly wider beam that is more forgiving of pointing choices made in advance. The difference is small ($\Delta w = 0.1$, or about 14% relative) but consistent.

14.4 Envelope-theorem sanity check at the optimum

Around $w^\star = 0.6$, nearby values give

$$V_{\text{adaptive}}(0.5) = 0.8522, \quad V_{\text{adaptive}}(0.6) = 0.8647, \quad V_{\text{adaptive}}(0.7) = 0.8532.$$

Central finite difference at $w = 0.6$:

$$\frac{V_{\text{adaptive}}(0.7) - V_{\text{adaptive}}(0.5)}{2 \cdot 0.1} = \frac{0.8532 - 0.8522}{0.2} = 0.005 \text{ nats/rad}.$$

Essentially zero, consistent with $w = 0.6$ being a (coarse-grid) interior maximum. A finer grid or an analytic envelope-theorem gradient evaluation would refine this.

14.5 Representative optimal sensor sequences at $w^\star$

The symmetry-reduced fixed-design search at $w = 0.6$ returns

$$s^\star_{\text{fixed}} = [1, 5, 9, 13],$$

four pointings equally spaced around the ring (quarter-rotations of each other). Any D_K-orbit representative is equivalent. This differs from the top-hat narrow-beam result $s^\star = [1, 7, 9, 13]$

in that the Gaussian-beam likelihood favors maximally-separated pointings (no two within a beamwidth of each other), whereas the top-hat was indifferent to separation as long as all four cells were distinct.

14.6 Takeaway: EVPI diagnoses, joint DP decides

The sweep is the cleanest possible demonstration of *why joint geometry-policy optimization is irreducible*. The four columns of Table 9—V_{oracle}, V_{adaptive}, V_{fixed}, and $\mathbb{E}[IG]$—behave qualitatively differently as w sweeps from 0 to π:

- V_{oracle} is monotonically decreasing. The oracle always benefits from a narrower beam because it knows where to point.
- V_{adaptive} has an interior maximum at $w = 0.60$ rad, value 0.865 nats.
- V_{fixed} has an interior maximum at $w = 0.70$ rad, value 0.758 nats. Different from both the above.
- $\mathbb{E}[IG] = V_{\text{oracle}} - V_{\text{fixed}}$ inherits the monotone decrease of V_{oracle} minus a shallow V_{fixed} curve: $\mathbb{E}[IG]$ is monotonically decreasing (on this grid).
- The *realized* adaptive advantage $V_{\text{adaptive}} - V_{\text{fixed}}$ has yet another maximum, at $w = 0.25$ rad (0.153 nats).

Consequently:

Necessary-but-not-sufficient claim: $\mathbb{E}[IG](\boldsymbol{c})$ is an upper bound on $V_{\text{adaptive}}(\boldsymbol{c}) - V_{\text{fixed}}(\boldsymbol{c})$, but is typically a *loose* bound, and choosing $\boldsymbol{c}$ to maximize $\mathbb{E}[IG]$ does *not* maximize the realized adaptive advantage. The two-regime comparison makes this stark:

$\boldsymbol{c}$	$\mathbb{E}[IG]$	$V_{\text{adaptive}} - V_{\text{fixed}}$	saturation
narrow top-hat ($m = 1$)	**2.081**	0.121	5.8%
wide top-hat ($m = K/2$)	0.513	**0.289**	**56.4**%
Gaussian $w = 0.6$	0.635	0.108	17.0%

The narrow design has the largest EVPI but the smallest realized gain; the wide top-hat has smaller EVPI but more than 2× the realized gain; and the Gaussian $w^\star$ maximum has the highest *absolute* V_{adaptive} but middling saturation. Three different $\boldsymbol{c}$ values are optimal under three different criteria, and no substitute for the joint DP will find the right $\boldsymbol{c}^\star$.

Use EVPI as the go/no-go gate. Use the joint DP to find the actual $(\boldsymbol{c}^\star, \pi^\star)$.

15 Related work and originality check

Before deploying this case study in a paper, we surveyed the adjacent literatures to check whether the specific problem formulation—finite-POMDP ring target search with envelope-theorem-based antenna-geometry co-design and an EVPI pre-solver gate—has been published elsewhere. The summary below is based on a targeted web search across cognitive-radar, sensor-management, adaptive-compressed-sensing, sequential Bayesian DoE, end-to-end-optics-co-design, programmable-metasurface, sequential-hypothesis-testing, and information-directed-sampling literatures.

15.1 Closest prior work in each community

Cognitive radar.

The closest published work is Bouhou et al., *POMDP-Driven Cognitive Massive MIMO*

Radar (arXiv:2410.17967, IEEE Trans. Radar Syst. 2025) and its follow-up *Joint Multi-Target Detection-Tracking via POMCP* (arXiv:2507.17506). These formulate cognitive MIMO radar as a POMDP but solve it approximately with Monte-Carlo tree planning (POMCP), adapt transmit-waveform and power, and *fix* the antenna aperture. Haykin's cognitive radar, Bell–Rangaswamy–Stinco, and Greco–Stinco–Gini's 2018 SPM survey all treat the aperture as given. No work in this community co-optimizes physical antenna element positions/excitations with an exactly-solved POMDP policy.

Sensor management.
The Hero–Cochran–Kreucher Foundations & Trends survey (2011) and the Hero–Castañón–Cochran–Kastella 2008 Springer volume cover POMDP sensor scheduling extensively (Kreucher–Kastella–Hero active sensing 2005; Castañón non-myopic classification). In every case the design variable is the scheduling policy over a fixed-menu sensor set; the physical hardware is never a co-design variable.

Adaptive compressed sensing.
Haupt–Castro–Nowak *Distilled Sensing* (IEEE Trans. Inf. Theory 2011) and Malloy–Nowak adapt *which rows* of an abstract sensing matrix to apply, without parameterizing the sensing matrix by physical array geometry and without POMDP DP.

Sequential Bayesian DoE.
Huan–Marzouk (arXiv:1604.08320, 2016) and Shen–Huan (arXiv:2110.15335, 2023) formulate sOED as a DP with continuous design spaces, solved approximately via ADP/policy-gradient RL. Alexanderian's 2021 *Acta Numerica* survey confirms the absence of exact Sondik-alpha-vector POMDP plus envelope-theorem physical-apparatus co-design in this community. The idea of EVPI / Expected Value of Sample Information *as a go/no-go gate* appears in clinical-trial VOI analysis (Flight–Julious–Brennan–Todd 2022) but not in radar/photonic sensor design.

End-to-end optics co-design.
Sitzmann et al. (SIGGRAPH 2018), Chang–Wetzstein 2019, Metzler–Heide–Wetzstein, Tseng–Majumdar, Kellman–Bostan–Waller (arXiv:1808.03571) jointly optimize static hardware with a *fixed, non-adaptive* reconstruction network or illumination sequence. No POMDP, no belief-state DP, no envelope-theorem policy/geometry split.

Programmable metasurfaces / RIS.
Compressive metasurface-aperture imaging (Yurduseven, Imani, Hunt, Smith) co-designs masks with reconstruction non-adaptively. RIS beamforming with deep RL uses Gaussian MIMO state models and abstract phase-shift layouts; hardware geometry is fixed.

Sequential hypothesis testing.
Chernoff (1959) and Naghshvar–Javidi *Active Sequential Hypothesis Testing* (Annals of Statistics 2013, arXiv:1203.4626) treat K-ary testing with adaptive experiment selection from a *fixed finite menu of actions.* The abstract structure matches our ring-search problem, but these papers do not parameterize the experiment's Fisher/Kullback matrix by continuous antenna geometry and do not identify closed forms like $\mathbb{E}[IG]_{\text{narrow}} = \frac{K-N}{K}\ln(K-N)$ or $\mathbb{E}[IG]_{\text{wide}} = \ln\big(K/(2N)\big)$.

Information-directed sampling.
Russo–Van Roy (JMLR 2018, arXiv:1403.5556) and recent work C-IDS on contextual POMDPs (arXiv:2602.03939) embed POMDPs in bilevel optimization, with abstract context variables and approximate policies—the closest *conceptual* match to our outer-context / inner-POMDP split—but not the envelope-theorem identity on the exact optimal value, and not the specific physical-geometry application.

15.2 What appears novel in the present framework

Individually well-established components: exact finite-POMDP DP via Sondik alpha-vectors (1971); adaptive radar beam scheduling with POMDPs; sequential Bayesian DoE as DP; envelope/Danskin theorem for bilevel optimization (Milgrom–Segal 2002); EVPI as a decision-theoretic quantity (classical Howard–Raiffa–Schlaifer); end-to-end differentiable hardware co-design (Wetzstein/Waller).

Combinations not found in the searched literature:

1. Treating antenna element positions and complex excitations as the outer design variable that *parameterizes the observation model of an exactly-solved finite POMDP*, with policy iteration alternating Sondik-exact Bellman and geometry gradient steps.
2. Using the envelope/Danskin identity on the optimal POMDP value so the geometry gradient reduces to the explicit partial at the fixed policy, avoiding implicit differentiation through the DP.
3. EVPI / ignorance gap as a pre-solver go/no-go gate with closed-form expectations $\mathbb{E}[IG]^{\text{narrow}}$ and $\mathbb{E}[IG]^{\text{wide}}$ under D_K-symmetric priors.
4. Orbit-symmetry (D_K dihedral) decomposition of the per-cell $IG(x)$ distribution for the ring search, yielding the two-value (narrow) and five-value (wide) histograms we observe numerically.

15.3 Caveat

Web-search coverage of paywalled radar venues (IET, IEEE AES, NATO STO) and thesis repositories is uneven, and classified cognitive-radar literature is not searchable. If a direct prior art exists it is likely in an obscure venue or a very recent unindexed preprint. We found no direct match across the surveyed communities, and the specific closed-form EVPI predictions and orbit-dependent per-cell $IG(x)$ distributions appear not to have been reported previously.

A Julia implementation

The following self-contained script produces all numerical values in this document. It is placed alongside this `.tex` file as `compute_ig.jl`.

```
const K = 16; const N = 4
const p_D = 0.9; const p_FA = 0.05
φ(z) = 2*pi * (z - 1) / K

function make_beam_coverage(m::Int)
    coverage = [falses(K) for _ in 1:K]
    half = div(m, 2)
    for a in 1:K, z in 1:K
        d = minimum((mod(z - a, K), mod(a - z, K)))
        coverage[a][z] = (m == 1 ? d == 0 :
                          (d <= half - 1) || (m % 2 == 1 && d == half))
    end
    coverage
end

p_detect(z, a, cov) = cov[a][z] ? p_D : p_FA

function posterior_entropy(s, y, cov)
    logb = zeros(K)
    for z in 1:K, k in 1:N
        pk = p_detect(z, s[k], cov)
        logb[z] += log(y[k] == 1 ? pk : 1 - pk)
    end
    m = maximum(logb); b = exp.(logb .- m); b ./= sum(b)
```

```
    -sum(b[z] > 0 ? b[z] * log(b[z]) : 0.0 for z in 1:K)
end

py_given_xs(y, x, s, cov) = prod(y[k] == 1 ? p_detect(x, s[k], cov) :
                                        1 - p_detect(x, s[k], cov) for k in 1:N)

function Φ_value(x, s, cov)
    lnK = log(K); val = 0.0
    for bits in 0:(2^N - 1)
        y = [((bits >> (k-1)) & 1) for k in 1:N]
        pyx = py_given_xs(y, x, s, cov)
        if pyx > 0
            val += pyx * (lnK - posterior_entropy(s, y, cov))
        end
    end
    val
end
```

The main loops (oracle and fixed enumerations) are straightforward four-nested loops over the pointings; see `compute_ig.jl` for the full listing.

B Sanity check: $\mathbb{E}[IG]$ as a scalar

As a consistency test,

$$\mathbb{E}[IG]^{\text{narrow}} = V_{\text{oracle}}^{\text{narrow}} - V_{\text{fixed}}^{\text{narrow}} = 2.579634 - 0.498683 = 2.080951$$

matches the independently-computed $\mathbb{E}_x[IG(x)] = 2.080950$ to five decimal places (the sixth is floating-point), confirming the arithmetic. Similarly in the wide regime:

$$1.923462 - 1.410526 = 0.512936 \;\approx\; 0.512935.$$